

Latents and Text

DATA 443/643 · Week 12, note 12.3

Week 12

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Ideas from Stanford CS231N 2025 L14 (LDM $D=8$, $C=16$; CFG). Original slides; that course is not copied.

Lecture plan

- 1 Denoise a VAE latent, not pixels
- 2 CLIP's text tower as c
- 3 Classifier-free guidance
- 4 $s=1$ is still conditional

Key goal: report s , a prompt-alignment score, and a failure case.

First time: what / why / formula

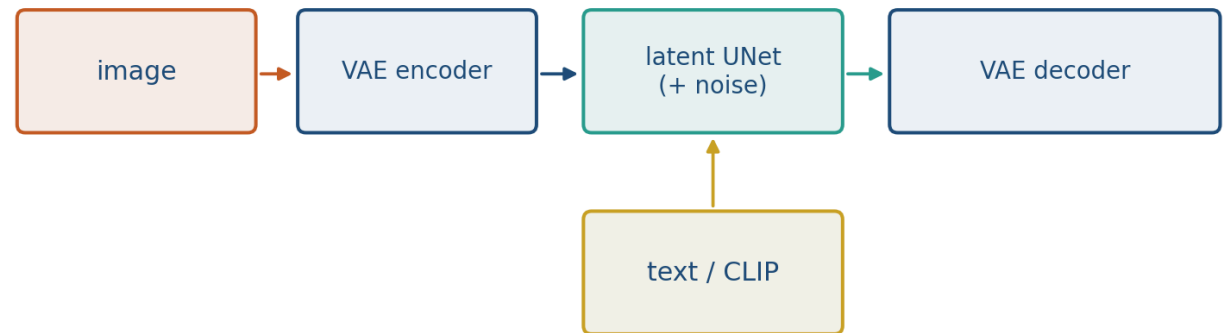
- What / why. Encode $x_0 \rightarrow z_0$ with a frozen VAE encoder, diffuse z , decode at the end. Text: condition on a CLIP (or T5) vector. CFG scale s sharpens the condition. Pixels are huge. Latent cells are fewer (on the order of $12 \times$ fewer in the Stable Diffusion cartoon). Text needs a condition or you sample generic images.
- Architecture. VAE enc/dec (frozen) + denoiser on z_t + text encoder. CFG: two forwards, $\hat{\epsilon} = \epsilon_u + s(\epsilon_c - \epsilon_u)$.
- Formula. $\hat{\epsilon}_\theta = \epsilon_\theta(z_t, t, \emptyset) + s(\epsilon_\theta(z_t, t, c) - \epsilon_\theta(z_t, t, \emptyset))$.
- Tradeoffs. + Cheaper than pixel DDPM, steerable with text. – VAE artifacts; CFG is a quality–diversity knob; still many reverse steps.

How. $s = 1$ is still conditional (not “off”). $s = 0$ is the uncond branch. High s : sharper, less diverse.

Compress, denoise, decode

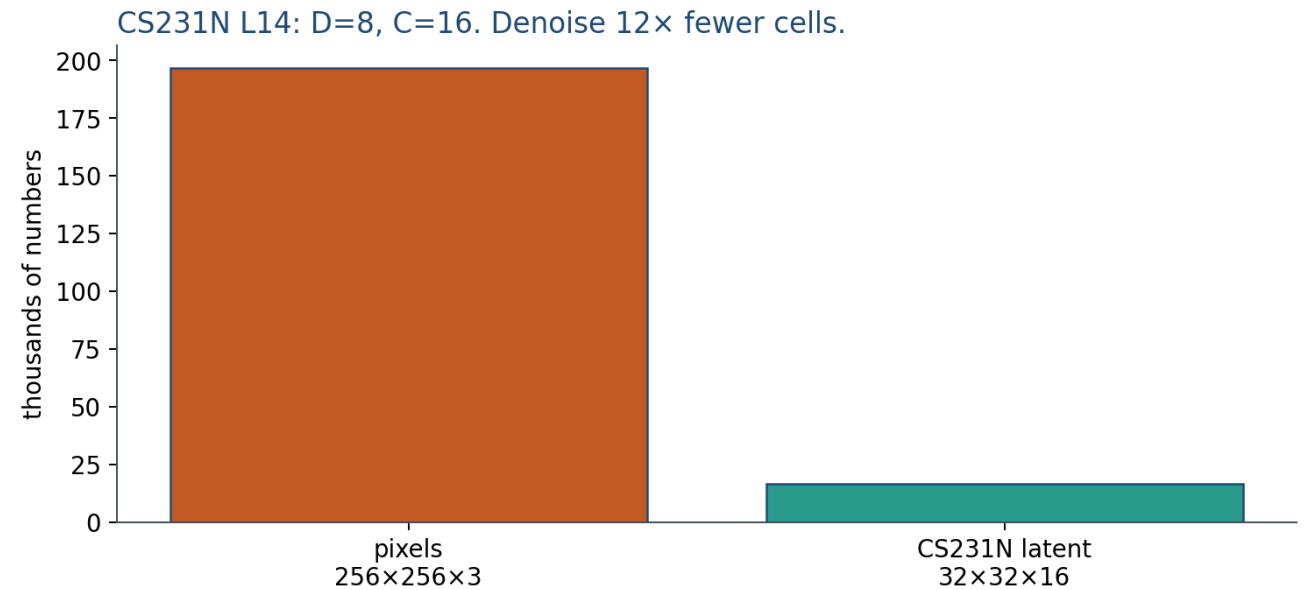
- Encoder frozen. UNet on z . Decoder back to pixels.
- A modern LDM stacks VAE + GAN + diffusion.
- DiT exists. This hour is the VAE latent picture.

Latent diffusion: denoise a cheap latent, condition on text



Count the cells

- $256 \times 256 \times 3$ vs $32 \times 32 \times 16$ ($D = 8$, $C = 16$).
- About $12\times$ fewer numbers per reverse step.
- Stable Diffusion's public recipe is $64 \times 64 \times 4$ on 512^2 : about $48\times$ fewer.



Classifier-free guidance

$$\hat{\varepsilon} = \varepsilon_u + s(\varepsilon_c - \varepsilon_u)$$

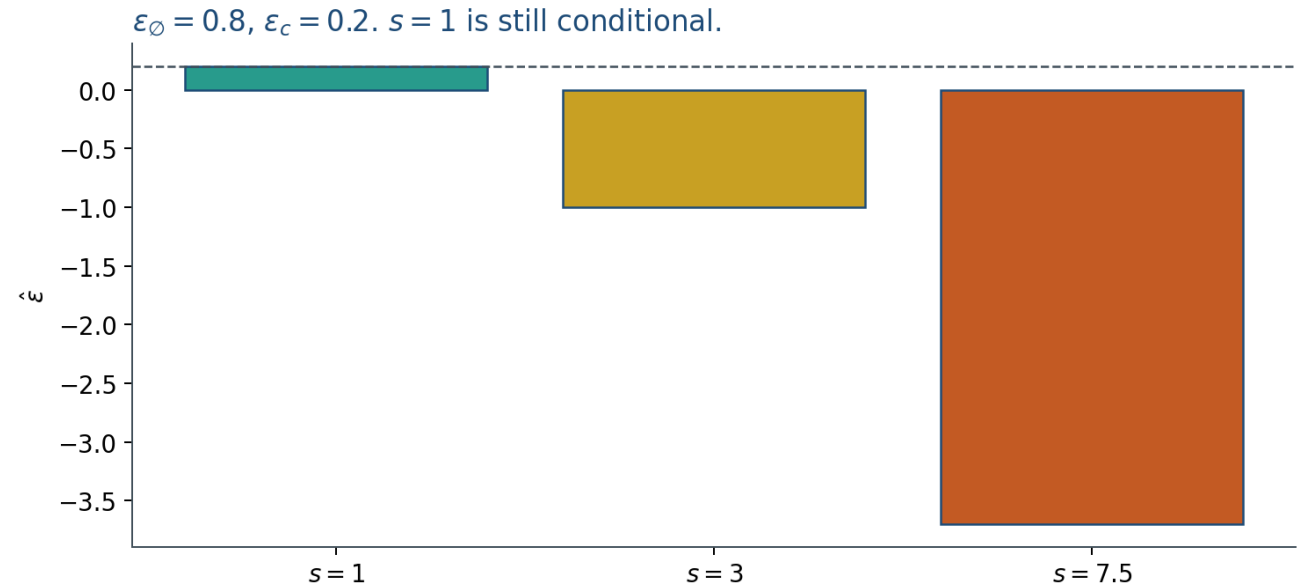
Train by randomly dropping the text.
One net, two calls.

$s = 1$ is the conditional prediction.
Not “no text.”

$s > 1$ extrapolates. Variety usually drops.

A scalar CFG

- $\varepsilon_u = 0.8, \varepsilon_c = 0.2$.
- $s = 1 \rightarrow 0.2$. $s = 3 \rightarrow -1.0$.
 $s = 7.5 \rightarrow -3.7$.
- CLIP's image tower scores pairs.
Sampling needs the text tower.



What to take from this lecture

Two calls

Empty prompt and text. Then mix with s.

Measure

CLIP cosine vs the prompt, plus a wrong-count failure.

Lab 12

Still 2-D denoising. No U-Net.